PEOPLE ANALYTICS · ENGINEERING · VOLUNTARY EXITS OVER 6 MONTHS · OCTOBER 2026

This model reliably found where Engineering's exit risk is highest — across 4 independent tests

22 groups
using data HR already has: time at the company, team exits and level; over 300 people per group

19.3% → 31.3%
the 5 highest-risk groups hold 19.3% of Engineers, but 31.3% of actual exits

3.3% vs 3.4%
estimated vs. actual exits, Jan–Jul 2026

How we tested it: At each date, we used only past data, predicted the next 6 months and checked actual exits. We repeated this 4 times, Feb 2024–Jan 2026. Informed by 43 studies; 4 model types tested. Before use: reduce overestimates in the highest-risk group and check fairness across groups.

From research to the pilot plan, in five steps

Each step supports the decision. All results come from Engineering data.

1

Research and selection

SLIDES 3–4

43 studies, search rules and 10 model types narrowed to one.

2

The chosen model

SLIDES 5–6

From monthly risk to a 6-month estimate for each group.

3

The groups it finds

SLIDE 7

Engineering's 5 highest-risk groups and how we chose them.

4

How well it works

SLIDE 8

Does it find higher-risk groups and estimate exits well? Results from 4 tests.

5

Next steps

SLIDE 9

Our plan for the next 2 weeks.

43 studies guided our choices; Engineering data shaped the final model

We favored repeatable methods, data tracked over time and actual exits.

9 **search themes**
Time to exit, turnover, technical talent, exit types, advanced models, HR factors, retention impact, fairness and text

▼

32 **logged searches**
Crossref, OpenAlex, publishers, JSTOR, arXiv, PubMed Central and repositories; Scholar for discovery

▼

5 **inclusion rules**
Turnover or retention, data over time, partial follow-up or changing risk, HR factors, or governance

▼

43 **studies, 1972–2026**
42 reviewed; 1 full text unavailable. Excluded duplicates, unclear methods and untested models.

What each research theme informed

Risk over time Monthly tracking, including stayers	5	Led us to monthly risk estimates
Workplace turnover Exits come in waves; validate vs. actual	4	Supported this approach for HR data
More complex models Benchmarked against simpler models	6	Small gains; kept as a benchmark
Why people leave Career, manager/team, engagement, ties	23	Grouped candidate signals and set their test order
Responsible use Privacy and fair treatment	5	Report by group and check fairness

The studies informed our approach. Every model was built and fitted with Qualcomm data.

We tested 4 of 10 model types and chose monthly risk

5 ruled out; 1 pending · criteria set Sep 30, 2026, before results · Engineering tests on slide 8

MODEL TYPE	STATUS	ESTIMATION ERROR vs. actual exits · lower is better	REASON
Monthly risk	✓	**6.1%**	**Uses each month (C1); lowest error (C2); best risk ranking (C3)**
6-month logistic model	✓	**14.5%**	Drops recent months (C1, C7); larger error (C2)
LightGBM	✓	**10.2%**	Harder to explain (C5); larger error (C2)
Historical average	✓	**7.1%**	Does not separate risk groups (C3)
Random survival forest	◐	—	Test pending
Cox model	✗	—	Less suited to monthly data (C1)
Exit types and transitions	✗	—	Exit types still need to be defined
Deep learning and text	✗	—	Hard to explain (C5); free text excluded (C6)
Impact of retention actions	✗	—	No recorded retention actions
HR relationship models	✗	—	Explains relationships, not exit risk (C3)

Estimation error: % difference between estimated and actual exits, averaged over 4 tests.

What we need from the model

- **C1** Monthly history
- **C2** Group-level estimates
- **C3** Finds high-risk groups
- **C4** Stable across dates
- **C5** Easy to understand
- **C6** Privacy & fairness
- **C7** Handles incomplete data
- **C8** Easy to update
- **C9** Backed by research

Error: comparison of base models using Engineering data, Oct 1, 2026.

8 signals and 2 combinations estimate monthly risk, then risk over 6 months

Monthly chance of a voluntary exit · final model as of Oct 3, 2026

1 · DATA

Monthly Engineering records

1,880,719 monthly person records, Jul 2021–Jul 2026. Each shows the known signals and any voluntary exit before the next record.

→

2 · MODEL

One statistical formula

Each signal adds to a score. A formula turns the total into a monthly probability from 0 to 1.

$$h = 1 - \exp(-\exp(\alpha + \Sigma\, \beta x))$$

→

3 · MONTHLY RISK

Risk changes through the year

The same group can have different risk in May and October. Month of year is one signal.

→

4 · RISK OVER 6 MONTHS

Monthly risks combine

Combine the 6 monthly risks. The same formula covers 3 or 12 months without rebuilding the model.

$$R_6 = 1 - (1-h_1)(1-h_2)...(1-h_6)$$

THE SIGNALS

Level
8 groups

Time at the company
playbook bands: <1, 1–2, 2–3, 3–5, 5–10, 10+ years

Month of year
seasonal patterns in exits

Team exits
voluntary exits in the current manager's team over 12 months: 0, 1, 2+

Also included
Manager changes · Direct reports · Time since promotion · Rehire · Level × tenure · Tenure × manager changes

New: playbook bands, promotion and rehire. Improved results in all 4 tests.

Source: Engineering analysis, Oct 1–3, 2026. Segment, job family and protected traits are excluded.

From monthly risk to a 6-month view for each group

Tables: Jan 25, 2026 snapshot; Feb–Jul risk; trained through Jul 2025 · Chart: Jul 26, 2026 snapshot · Engineering

Step 1 Monthly risk for each person · fictional examples

PERSON AND GROUP	FEB	MAR	APR	MAY	JUN	JUL	R_6 FEB–JUL
I **Fernando** 3–5 years · level 8 · 2+ team exits	2.1%	3.6%	2.3%	1.7%	4.2%	2.6%	**15.5%**
II **Leah** 5–10 years · level 7 · no team exits	0.6%	1%	0.6%	0.5%	1.2%	0.7%	**4.5%**
III **Carlos** 10+ years · level 6 · no team exits	0.2%	0.3%	0.2%	0.1%	0.4%	0.2%	**1.4%**

Fernando: $R_6 = 1 - (1-2.1\%)(1-3.6\%)(1-2.3\%)(1-1.7\%)(1-4.2\%)(1-2.6\%) \approx 15.5\%$

Step 2 Monthly average risk for each group

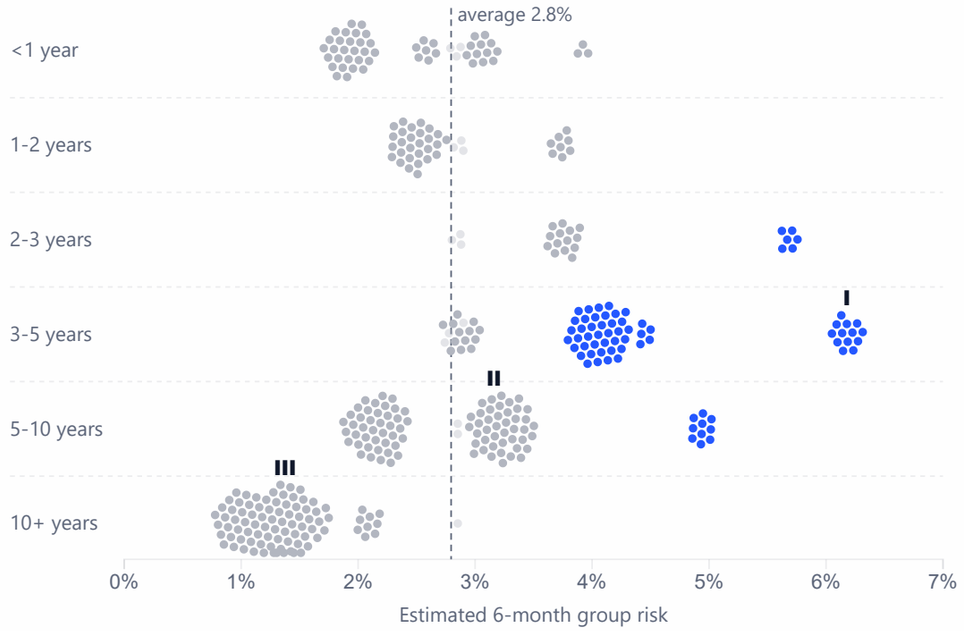
GROUP	FEB	MAR	APR	MAY	JUN	JUL	R_6 FEB–JUL	ACTUAL
I **3–5 yrs · team exits · levels 7–8** 1,373 people at snapshot	1%	1.6%	1%	0.8%	1.9%	1.2%	**7.2%**	**5.5%**
II **5–10 yrs · no exits · levels 7–8** 2,515 people at snapshot	0.5%	0.8%	0.5%	0.4%	1%	0.6%	**3.8%**	**4%**
III **10+ yrs · no exits · levels 5–6** 3,972 people at snapshot	0.2%	0.3%	0.2%	0.2%	0.4%	0.3%	**1.6%**	**1.5%**
All Engineering 34,963 people at snapshot	0.4%	0.7%	0.5%	0.3%	0.9%	0.5%	**3.3%**	**3.4%**

Group I: 7.2 exits estimated per 100 people, Feb–Jul; 5.5 actually left. Levels 7–8 are overestimated (slide 8).

Step 3 How we form the groups · each dot = 100 people

All | + Years at Qualcomm | + Team exits | + Level

● 5 highest-risk groups ● other groups · faint: groups under 300 people



22 groups

6.2% highest group risk

91% of individual risk ranking kept (automatic grouping: 85%)

Fictional names and profiles; risks come from the real model. Group R_6 combines monthly averages. Actual rates shown with 30+ exits. Dots sit around group averages, not individual risk. This grouping kept the most of the model's ranking across 58 options. Source: Engineering analysis, Oct 1–3, 2026.

The top 5 risk groups held 19.3% of people and 31.3% of actual exits

Estimated % leaving within 6 months · Jul 26, 2026 snapshot · 37,221 people · grouped by years at Qualcomm, team exits and level

	Years at Qualcomm	Team exits	Level	Estimated 6-month risk · line: average 2.8%	People	Latest test estimated · actual
1	3-5 years	With exits	Levels 7–8	6.2%	1,310 3.5%	6.9% · 5.5%
2	2-3 years	With exits	Levels 7–8	5.7%	572 1.5%	6.5% · 4.8%
3	5-10 years	With exits	Levels 7–8	4.9%	1,028 2.8%	5.4% · 3.8%
4	3-5 years	With exits	Levels 5–6	4.5%	478 1.3%	4.7% · 5.4%
5	3-5 years	No exits	Levels 7–8	4.1%	3,807 10.2%	4.6% · 3.9%

Top 5 of 22 groups; the other 17 have lower risk. Four have recent team exits; 4 of 5 are levels 7–8.

31.3%

of actual exits in the latest test, with 19.3% of people; ranking individuals would capture 31.9%

Levels in each band

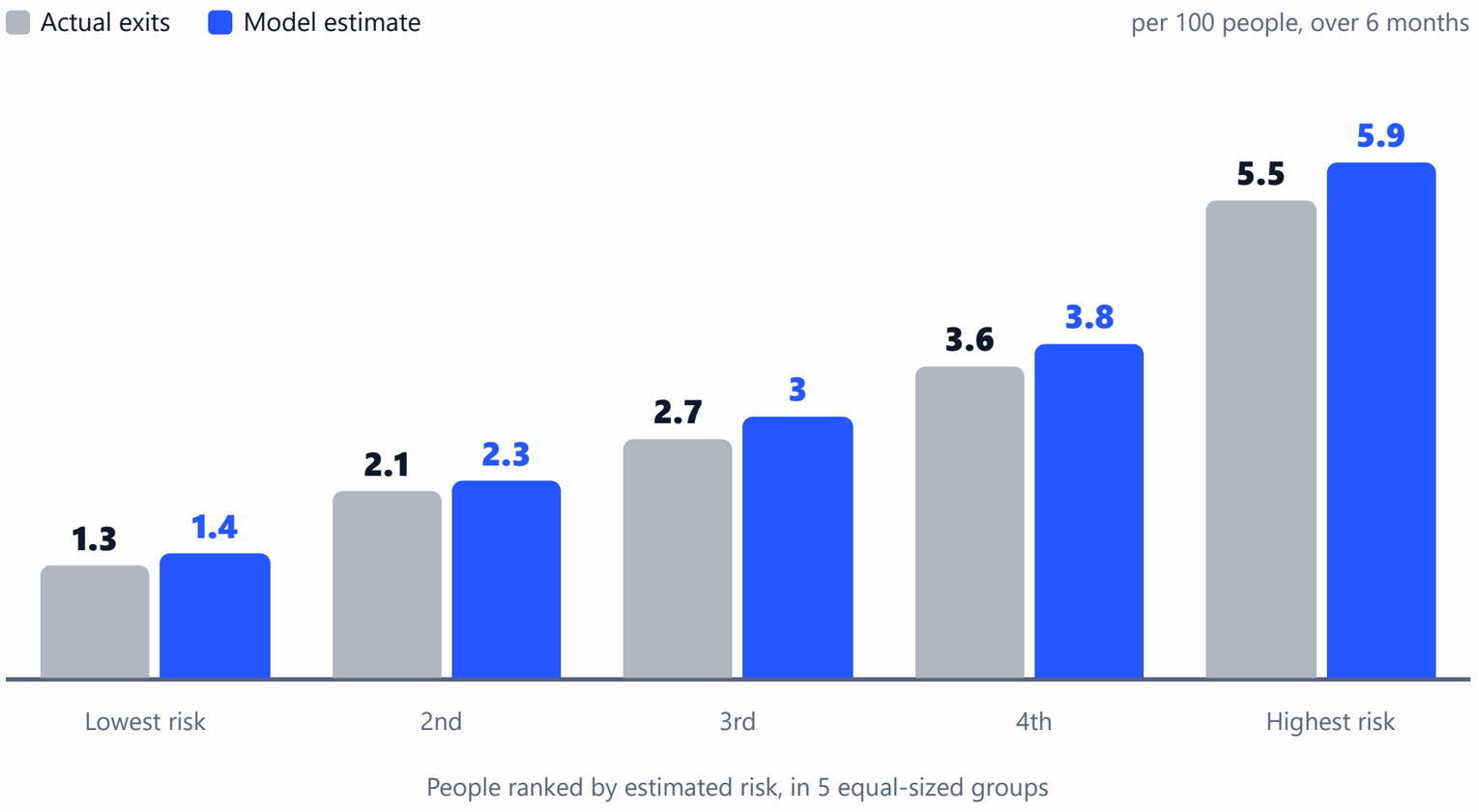
Levels 7–8	8 - Assoc & Int/Eng 7 - Senior & Lead
Levels 5–6	6 - Staff & /Mgr 5 - Sr Staff & /Mgr
Levels 1–3	3 - Princ/Directors 2 - Sr Dir 1 - VP & Above

Orange: estimate is 1 percentage point or more above actual. The model overestimates levels 7–8 (slide 8).

Chosen from 58 options: keeps 91% of the model's risk ranking across 4 tests (automatic grouping: 85%). Team exits = 1+ resignations in the current manager's team over 12 months; 2+ are linked to +86% monthly risk. One group per person; 300+ people and 30+ exits in the latest test. Source: Engineering analysis, Oct 1–3, 2026.

The highest-risk group had 4 times the exits of the lowest; estimates ran slightly high

4 tests on unseen data, Feb 2024–Jan 2026 · exits per 100 people over 6 months · Engineering



1 · Finds higher-risk groups

Per 100 people, 1.3 left in the lowest-risk group and 5.5 in the highest. Engineering's average is 3.

2 · Estimates are close, but high

Estimates ran high in every group: 4% to 11% above total exits. The highest-risk group has the largest gap and is our first fix.

Even in the highest-risk group, 95 of 100 stay. Use the model to choose groups for support, not individuals.

Each group has 30+ exits in every test. Each model used only earlier data. Source: Engineering analysis, Oct 1–3, 2026.

Next steps: the next 2 weeks

Improve, fix or replace the model before use

1. **Test the remaining model**
   Random survival forest is the only model family still untested
2. **Check more signals**
   QVoice (intent to stay) and time at current level
3. **Fix and validate**
   Reduce overestimates in the highest-risk group and check fairness across groups
4. **Choose the final version**
   Use the same 4 tests; adopt changes only if they improve today’s model

Then pilot the final model in 1 or 2 groups and update it regularly.